ROBERT DYRO

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PROFILE. My expertise is in computational engineering and applied machine learning. My doctoral work in Robotics at Stanford University provided me with a deep theoretical understanding and practical experience building systems combining applied machine learning and real-time optimization—work which I published at top statistical and robotics conferences.

As part of the JAX team, I have been focused on developing high-performance inference, serving and training of LLMs. This included both high-level distributed computing and low-level kernel development. I have advised and directly worked with production and research teams at Google and Google DeepMind on TPU and GPU performance.

TECHNICAL SKILLS. Programming Languages: *Python, C++*; Scientific Computation: *Julia, Matlab, Maxima*; ML/Deep Learning: *Deep knowledge of the JAX/Pallas kernels/XLA compiler stack on TPUs and GPUs*, several years of experience with *PyTorch on GPUs, Triton, CUDA*; Systems & Tools: *Slurm, GCP.*

CITIZENSHIP. Polish Citizenship (STEM OPT until 2027-06 or another work visa).

EDUCATION

Stanford University Stanford, CA
PhD, Robotics 2020 - 2024
MS, Aeronautics & Astronautics Engineering 2018 - 2020

University of California, Los Angeles Los Angeles, CA
BS, Aerospace Engineering, Minor in Philosophy, Summa Cum Laude 2014 - 2018

WORK HISTORY

Software Engineer, Google 2024 - Present, Mountain View, CA
On the JAX team I have focused on developing high-performance implementations of LLM models. In addition to core JAX development, I created and maintain high-performance JAX LLM examples including TPU and GPU implementations of Deepseek R1, GPT OSS, Kimi K-2, Qwen 3 MoE and others. I also implemented a lightweight serving framework in JAX, achieving full GPU and TPU utilization—matching the top commercial implementations. This lead me to consult and collaborate with production teams at Google and research teams at Google DeepMind. On production teams, I was one of the few lead kernel performance engineers, working on Pallas TPU kernels for the newest generation of Google's TPU chips (Ironwood). With research teams (both inside Google DeepMind and outside at e.g., Stanford, Berkeley, Apple), I consulted on LLM performance optimizations, helping to accelerate research.

PhD Student, Autonomous Systems Laboratory (ASL) at Stanford University 2020 - 2024, Stanford, CA
My PhD focused on machine learning and optimization in robotics and autonomous vehicle applications. I authored novel real-time optimization and machine learning techniques and focused on consensus optimization, meta-learning (learning through optimization), bilevel programming with differentiable optimization and numerical linear algebra sketching for analysis of machine learning behavior prediction networks.

Research Intern, Cruise 2022, San Francisco, CA
As a research intern, I worked on Zero-Shot Neural Architecture Search (NAS) internal company research and developed its original production NAS framework.

Research Intern, Toyota Research Institute 2020, Los Altos, CA
I interned at the self-driving research team and researched human logic embedding—via Signal Temporal Logic (STL)—to computationally capture human-interpretable behavior specification in learned behavior models.

TECHNICAL EXPERIENCE

Relevant Coursework: During my time at Stanford, I took courses focusing on Machine Learning (Meta-learning, Explainable AI, RL, Graph ML, Distribution Shift ML) and Optimization (Convex Optimization, Optimal Control Theory).

My other technical experience includes developing a library for automatic kernel tuning in JAX and Pallas; a differentiable
Pytorch-JAX interoperability layer. My recent experimental projects included a custom quadratic program (QP) solver in CUDA; an experimental tracing-based auto-differentiation library with sparse linear algebra support.

SELECTED PUBLICATIONS

- Dyro, R. (2024). "Optimization-Driven Autonomous Driving: Control, Intent, and Testing." *PhD diss., Stanford University.*
- Dyro, R., Foutter, M., Li, R., Di Lillo, L., Schmerling, E., Zhou, X., and Pavone, M. (2024). "Realistic extreme behavior generation for improved AV testing." *International Conference on Robotics and Automation (ICRA).*
- Dyro, R., Schmerling, E., Arechiga, N., and Pavone, M. (2022). "Second-order sensitivity analysis for bilevel optimization." *International Conference on Artificial Intelligence and Statistics (AISTATS).*
- Dyro, R., Harrison, J., Sharma, A., and Pavone, M. (2021). "Particle MPC for uncertain and learning-based control." *International Conference on Intelligent Robots and Systems (IROS).*

MISC

Philosophy Minor ■ LA Marathon ■ Ham Radio ■ PADI Assistant Instructor ■ I speak English, Polish & German.